Dosing Frequency – Example 1

Order: Amoxicillin 1,500 mg/day in divided doses
Frequency: TID (3 doses per day)
Question: How many mg should be given per dose?

Step 1: Identify the total daily dose and the number of doses per day.

Step 2: Write the total daily dose as a fraction over 1 day.

Step 3: Write the frequency as a conversion factor (day $\rightarrow$ doses).

Step 4: Multiply so that “day” cancels.

Step 5: Calculate and write the dose per administration.

Dosing Frequency – Example 1 Solution

Order: Amoxicillin 1,500 mg/day in divided doses
Frequency: TID (3 doses per day)
Question: How many mg should be given per dose?

Step 1: Identify the total daily dose and the number of doses per day.

Total daily dose: $1{,}500$ mg/day   Frequency: 3 doses/day

Step 2: Write the total daily dose as a fraction over 1 day.

$$\frac{1{,}500 \text{ mg}}{1 \text{ day}}$$

Step 3: Write the frequency as a conversion factor (day → doses).

$$\boxed{\frac{1 \text{ day}}{3 \text{ doses}}}$$

Step 4: Multiply so that "day" cancels.

$$\frac{1{,}500 \text{ mg}}{1 \cancel{\text{day}}} \times \boxed{\frac{1 \cancel{\text{day}}}{3 \text{ doses}}} = \underline{\qquad} \frac{\text{mg}}{\text{dose}}$$

Step 5: Calculate and write the dose per administration.

$$\frac{1{,}500}{3} = 500 \frac{\text{mg}}{\text{dose}}$$

Administer 500 mg per dose.

Order: Metformin 2,000 mg/day in divided doses
Frequency: BID (2 doses per day)
Question: How many mg should be given per dose?

Step 1: Identify the total daily dose and the number of doses per day.

Step 2: Write the total daily dose as a fraction over 1 day.

Step 3: Write the frequency as a conversion factor (day $\rightarrow$ doses).

Step 4: Multiply so that "day" cancels.

Step 5: Calculate and write the dose per administration.

Dosing Frequency – Example 2 Solution

Order: Metformin 2,000 mg/day in divided doses
Frequency: BID (2 doses per day)
Question: How many mg should be given per dose?

Step 1: Identify the total daily dose and the number of doses per day.

Total daily dose: $2{,}000$ mg/day   Frequency: 2 doses/day

Step 2: Write the total daily dose as a fraction over 1 day.

$$\frac{2{,}000 \text{ mg}}{1 \text{ day}}$$

Step 3: Write the frequency as a conversion factor (day $\rightarrow$ doses).

$$\boxed{\frac{1 \text{ day}}{2 \text{ doses}}}$$

Step 4: Multiply so that "day" cancels.

$$\frac{2{,}000 \text{ mg}}{1 \cancel{\text{day}}} \times \boxed{\frac{1 \cancel{\text{day}}}{2 \text{ doses}}} = \underline{\hspace{2em}} \frac{\text{mg}}{\text{dose}}$$

Step 5: Calculate and write the dose per administration.

$$\frac{2{,}000}{2} = 1{,}000 \frac{\text{mg}}{\text{dose}}$$

Administer 1,000 mg per dose.

Dosing Frequency – Example 3

Order: Penicillin V 1,000 mg/day in divided doses
Frequency: QID (4 doses per day)
Question: How many mg should be given per dose?

Step 1: Identify the total daily dose and the number of doses per day.

Step 2: Write the total daily dose as a fraction over 1 day.

Step 3: Write the frequency as a conversion factor (day $\rightarrow$ doses).

Step 4: Multiply so that "day" cancels.

Step 5: Calculate and write the dose per administration.

Dosing Frequency – Example 3 Solution

Order: Penicillin V 1,000 mg/day in divided doses
Frequency: QID (4 doses per day)
Question: How many mg should be given per dose?

Step 1: Identify the total daily dose and the number of doses per day.

Total daily dose: $1{,}000$ mg/day   Frequency: 4 doses/day

Step 2: Write the total daily dose as a fraction over 1 day.

$$\frac{1{,}000 \text{ mg}}{1 \text{ day}}$$

Step 3: Write the frequency as a conversion factor (day $\rightarrow$ doses).

$$\boxed{\frac{1 \text{ day}}{4 \text{ doses}}}$$

Step 4: Multiply so that "day" cancels.

$$\frac{1{,}000 \text{ mg}}{1 \cancel{\text{day}}} \times \boxed{\frac{1 \cancel{\text{day}}}{4 \text{ doses}}} = \underline{\qquad} \frac{\text{mg}}{\text{dose}}$$

Step 5: Calculate and write the dose per administration.

$$\frac{1{,}000}{4} = 250 \frac{\text{mg}}{\text{dose}}$$

Administer 250 mg per dose.